
بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ



What is Natural Language Processing

What is Natural Language

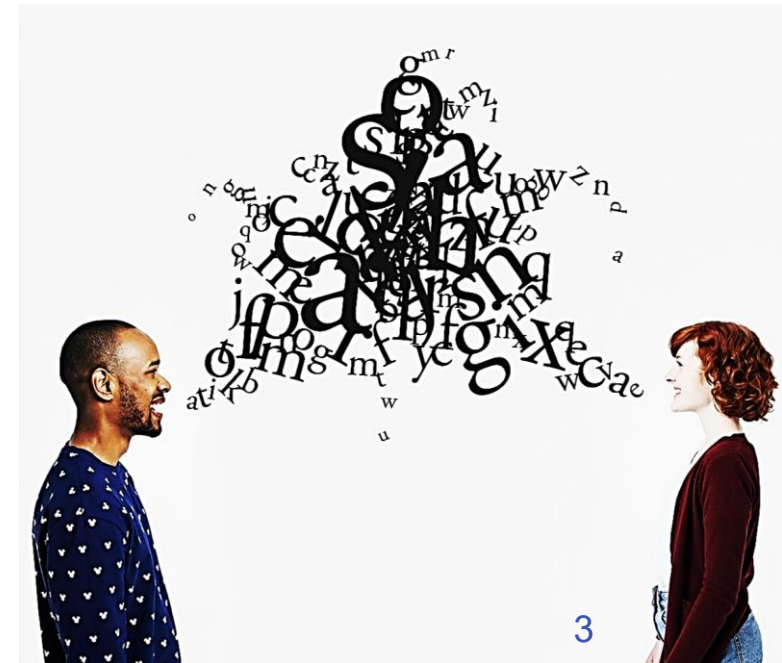
- ❖ **Natural language** refers to the language spoken, written, or signed by humans for general communication (e.g., English, Urdu, Spanish). Unlike artificial languages (like Python or C++), natural languages evolve naturally within human communities over centuries and lack strict, unambiguous mathematical rules

For example

- "I'm down for that"
- "I'm up for that."

Math Logic: "Up" and "Down" are opposites.

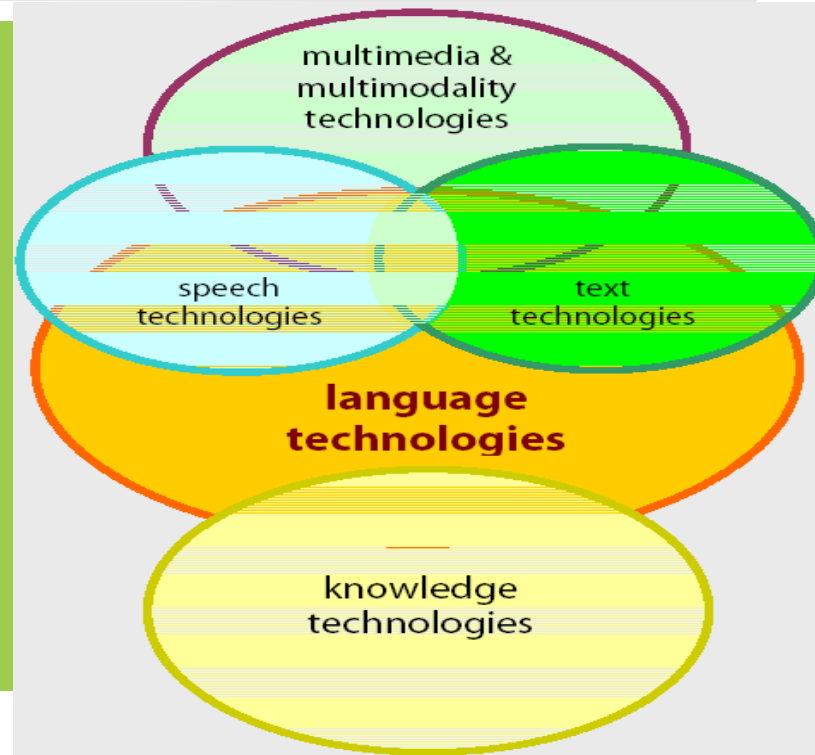
Human Logic: Both expressions mean the same thing.



NLP – Ultimate Goal



To develop such a **System** which behaves like Human



What is NLP

- ❖ **Wiki: Natural language processing (NLP)** is a field of computer science, artificial intelligence, and computational linguistics concerned with the interactions between computers and human (**natural**) languages.



Parts of NLP

Term	Definition	Example	Application
NLP (Natural Language Processing)	Field dealing with the interaction between computers and human languages.	Text classification (spam detection), translation (Google Translate).	Chatbots, search engines, voice assistants.
NLU (Natural Language Understanding)	Focuses on understanding and interpreting the meaning and intent behind natural language.	Virtual assistants (Siri), sentiment analysis.	Customer service automation, intent recognition.
NLG (Natural Language Generation)	Generates natural language from structured data.	Generating weather reports, automatic text summarization.	Report generation, content creation, chatbots.

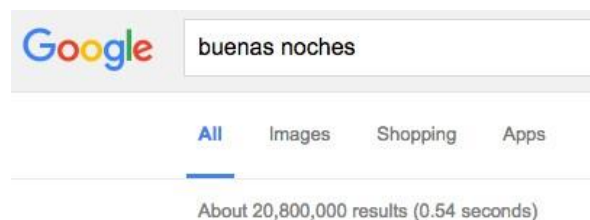
Go beyond the keyword matching



- ❖ Identify the **structure** and **meaning** of **words**, **sentences**, **texts** and **conversations**
- ❖ **Deep** understanding of **broad** language
- ❖ NLP is all around us

Applications

Machine Translation



Open in Google Translate



Facebook translation, image credit: Meedan.org

Dialog Systems

Gift shop

Items such as caps, t-shirts, sweatshirts and other miscellanea such as buttons and mouse pads have been designed. In addition, merchandise for almost all of the projects is available.



Hi. I'm your automated online assistant. How may I help you?

CD or DVD

There is a series of CDs/DVDs with selected Wikipedia content being produced by Wikipedians and [SOS Children](#).

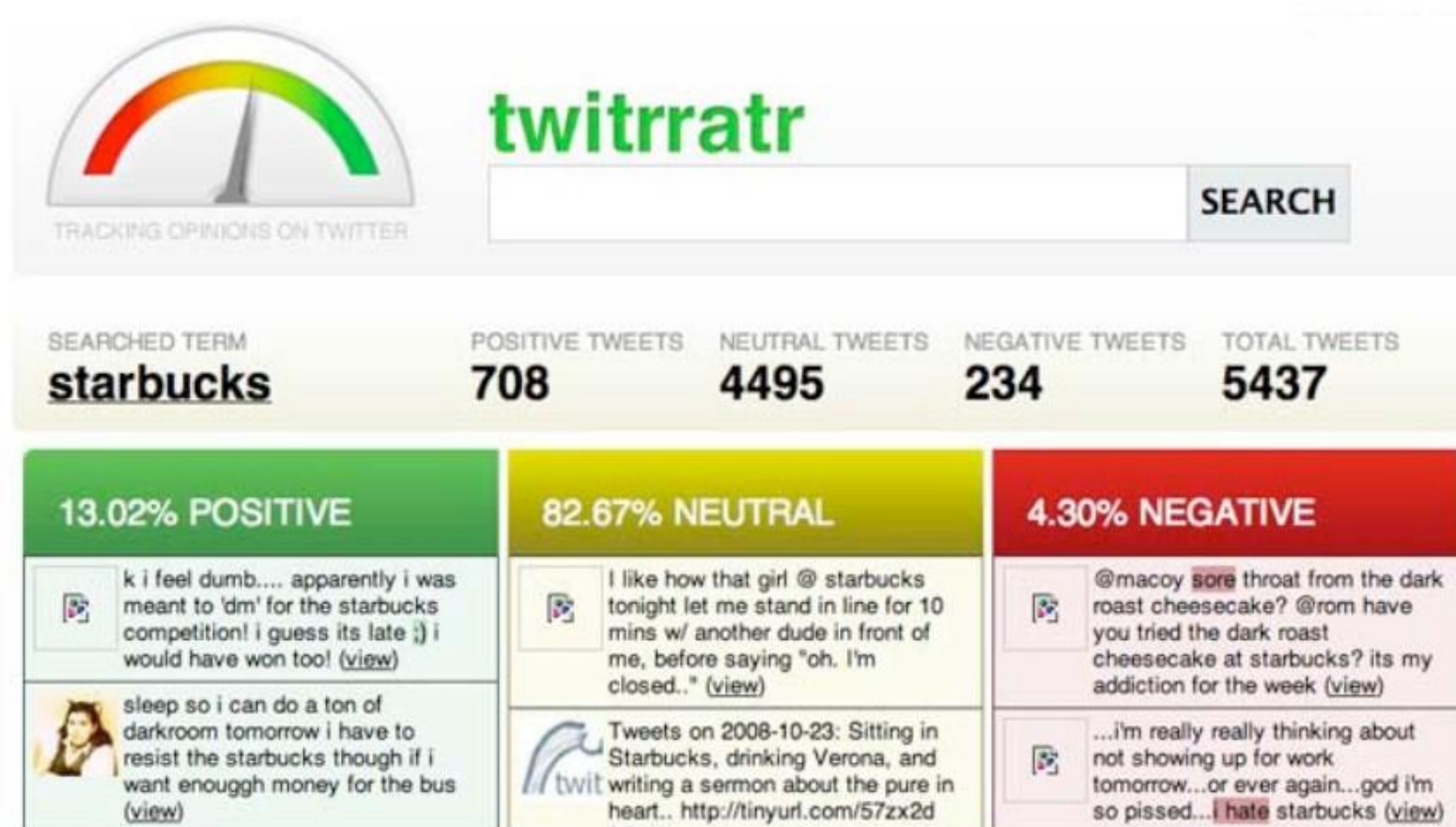


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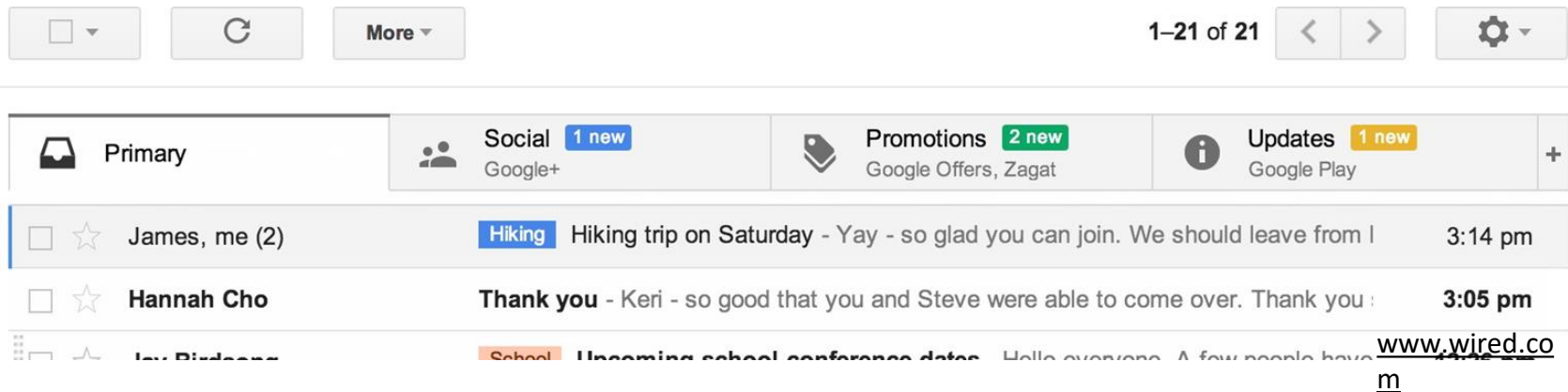
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Sentiment/Opinion Analysis



Text Classification

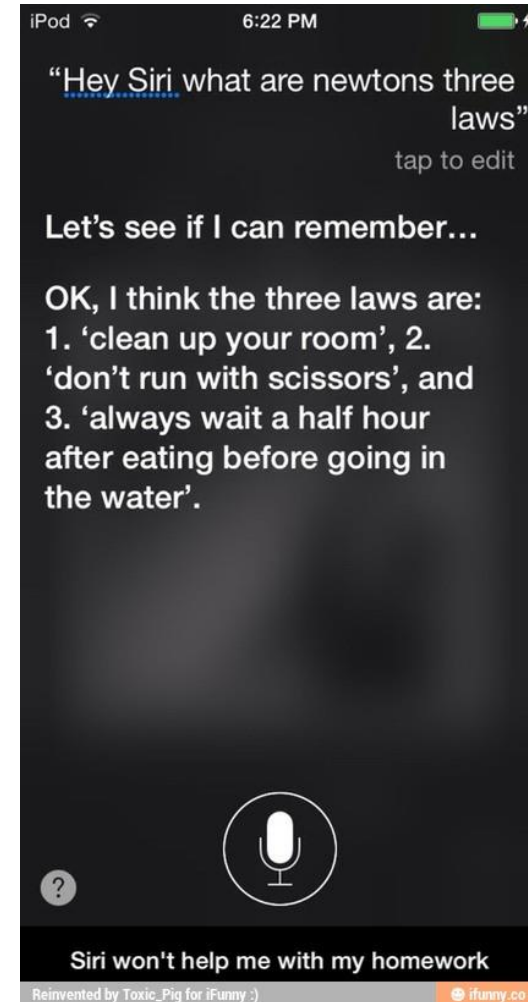


❖ Other applications?

Question answering



'Watson' computer wins at 'Jeopardy'



credit:
ifunny.com

Question answering

❖ Go beyond search

What's the capital of Wyoming?

Web Maps Shopping Images News More ▾ Search tools

About 984,000 results (0.54 seconds)

Wyoming / Capital



Cheyenne

Information Extraction

❖ Unstructured text to database entries

New York Times Co. named Russell T. Lewis, 45, president and general manager of its flagship New York Times newspaper, responsible for all business-side activities. He was executive vice president and deputy general manager. He succeeds Lance R. Primis, who in September was named president and chief operating officer of the parent.

Person	Company	Post	State
Russell T. Lewis	New York Times newspaper	president and general manager	start
Russell T. Lewis	New York Times newspaper	executive vice president	end
Lance R. Primis	New York Times Co.	president and CEO	start

Yoav Artzi: Natural language processing

Language Comprehension

Christopher Robin is alive and well. **He** is the same person that you read about in the book, **Winnie the Pooh**. As a boy, **Chris** lived in a pretty home called **Cotchfield Farm**. When **Chris** was three years old, **his father** wrote a poem about **him**. The poem was printed in a magazine for others to read. **Mr. Robin** then wrote a book

- ❖ Q: who wrote Winnie the Pooh?
- ❖ Q: where is Chris lived?

More Applications

Speech Recognition

"To convert audio into text for further processing"

to convert audio into text for further processing.

Automatic Summarization

"To add summaries to existing documents"

Chatbots

"To provide 24/7, 365 days a year customer support"

Customer Support without human support

Video Captioning

"To extract and analyse information from videos in the form of text"

Image Captioning

"To annotate images"

More Applications...

Spell Checking

"To make sure documents don't contain errors"

to make sure that documents don't contain errors. Especially relevant in industries with high compliance requirements such as banking, insurance and finance.

Advertisement Matching

"To match a user's online text input or queries with relevant ads"

For example, when you search for a new car on Google or write an email about it to a friend, you will later receive car adverts.

Keyword Search

"To locate relevant information faster across all data facilities"

to locate relevant information faster across all data facilities, leading to increased efficiency.

Text Generation

"To create new text, e.g. client contracts, research documents, etc."

to create new text, e.g. client contracts, research documents, training materials, and so on.

Top NLP Tools: Libraries & Frameworks

NLTK (Natural Language Toolkit)

SpaCy

Gensim

TensorFlow and PyTorch

Scikit-learn

Tools [Libraries & Frameworks] For Large Language Models (LLMs)

[Hugging Face](#) Transformers

LangChain

LlamaIndex

Haystack

NLP Cont...



Without NLP / Machine Learning, we cannot develop intelligent programs which can **assist** humans in a **range of tasks**

How does NLP work...

- NLP works by breaking down unstructured human language into structured data that a computer can process.
- Morphology: What is a word?
 - Morphology deals with the structure and formation of words. It studies how words are formed from morphemes, which are the smallest grammatical units (like prefixes, roots, and suffixes).
 - **Example**: The word "unhappiness" is made up of three morphemes: "un-" (prefix), "happy" (root), and "-ness" (suffix).

- Lexicography: What does each word mean?

- He plays bass guitar.
- That bass was delicious!



Word Sense Disambiguation (WSD) in NLP deals with figuring out the correct meaning of a word based on the context

- Syntax: How do the words relate to each other? (the arrangement of words in a sentence and the rules governing sentence structure.)
 - The dog bit the man. ≠ The man bit the dog.

NLP • But in Russian: человек собаку съел = человек съел собаку

Language Technology

making good progress

mostly solved

Spam detection

Let's go to Agra!



Buy V1AGRA ...



Part-of-speech (POS) tagging

ADJ ADJ NOUN VERB ADV

Colorless green ideas sleep furiously.

Named entity recognition (NER)

PERSON ORG LOC

Einstein met with UN officials in Princeton

Sentiment analysis

Best roast chicken in San Francisco!



The waiter ignored us for 20 minutes.

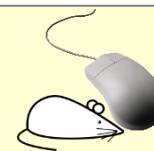


Coreference resolution

Carter told Mubarak he shouldn't run again.

Word sense disambiguation (WSD)

I need new batteries for my *mouse*.



Parsing

g

I can see Alcatraz from the window!

Machine translation (MT)

第13届上海国际电影节开幕...



The 13th Shanghai International Film Festival...

Information extraction (IE)

You're invited to our dinner party, Friday May 27 at 8:30



Party
May 27
add

still really hard

Question answering (QA)

Q. How effective is ibuprofen in reducing fever in patients with acute febrile illness?

Paraphras

XYZ acquired ABC yesterday

ABC has been taken over by XYZ

Summarization

The Dow Jones is up

The S&P500 jumped

Housing prices rose



Economy is good

Dialog

Where is Citizen Kane playing in SF?



Castro Theatre at 7:30. Do you want a ticket?



Why is NLP difficult?

- **Word category ambiguity**

book --> verb? or noun?

Explanation: Many words in a language can belong to multiple grammatical categories (parts of speech), and their category is determined by context.

The word “**book**” can function as different parts of speech.

As a Noun (thing/object)

*“I read a **book**.”*

Here, **book** refers to a physical object used for reading.

As a Verb (action)

*“I will **book** a ticket.”*

Here, **book** means **to reserve or make a reservation**.

Challenge for NLP: The system needs to determine whether "book" is being used as a noun or a verb based on the surrounding context.

Why is NLP difficult?

- **Word sense ambiguity**

bank --> financial institution? building? or river side?

A single word can have multiple meanings (senses), and distinguishing between these meanings requires understanding the context in which the word is used.

Example: The word "bank" can refer to:

A financial institution: "I deposited money in the bank."

A riverbank: "We walked along the bank of the river."

Challenge for NLP: The system must perform Word Sense Disambiguation (WSD) to correctly identify the intended meaning of "bank" in different contexts.

Why is NLP difficult?

Different more challenges in Natural Language Processing (NLP)

1. Words Can Mean More Than the Sum of Their Parts

Sometimes the meaning of a phrase cannot be understood by simply combining the meanings of individual words.

Example:

“Make up a story.” -> make = create, up = direction/position, story = narrative, But together “make up” means to invent

2. Defining Scope (Quantifier Ambiguity)

Sometimes a sentence is unclear about how widely a statement applies.

Example: “People like ice-cream.”

Possible interpretations: All people like ice-cream., Most people like ice-cream., Some people like ice-cream.

The sentence does not clearly define the scope of “people.”

3. Language is Changing and Evolving

Language constantly changes with technology, culture, and new words.

Example: “I’ll email you my answer.”

Originally, email was only a noun (electronic mail).

Now it is also used as a verb meaning to send a message electronically.

➡ Words can change grammatical roles over time.

4. Structural or Grammatical Issues

Example: “This new S.U.V. has a compartment for your mobile phone.”

Here, the meaning is clear to humans, but NLP systems must identify:

S.U.V. as a vehicle, compartment as storage space, mobile phone as a device

The system must understand the relationship between objects in the sentence.

NLP Cont...



When to Use?

- When we have **Data** with three main characteristics

- I. Large amount of Data
- II. High quality Data
- III. Balanced Data

- When **patterns** exist in our Data
 - Even if we don't know what they are?

Major Challenges in NLP Cont...

However, we **can build intelligent programs**, which can do several useful tasks like

1	Spam Email Detection	7	Next Word Prediction
2	Gender Identification	8	Text Summarization
3	Age Group Identification	9	Speech to Text
4	Sentiment Analysis	10	Text to Speech
5	Face Recognition	11	Hate Speech Classification
6	Machine Translation		



Steps to Build Efficient NLP/ Machine Learning Models

Steps to Build **Efficient** NLP / Machine Learning Models

- 1** Build **strong** and **accurate** understanding of Data
- 2** **Properly** pre-process Data, so that it becomes **high-quality data**
- 3** **Represent / Transform** Data into a format which Machine Learning Algorithms can understand

Steps to Build **Efficient** NLP / Machine Learning Models

4

Identify **what** Machine Learning Algorithms will be **most suitable** for your Data (prepared in Step 3)

5

Train / Test **selected** (in Step 4) Machine Learning Algorithms on Data (prepared in Step 3)



Treating Real World Problems as Learning Input-Output Functions

Example 01 – Treating Gender Identification Problem as Learning Input-Output Function

Available Data

Instance No.	Input			(Output)
	Height	Weight	Beard	Gender
d_1	Short	Medium	No	Female
d_2	Tall	Heavy	Yes	Male
d_3	Medium	Medium	No	Female

Example 02 – Treating Gender Identification Problem as Learning Input-Output Function

Available Data

Instance No.	Input (Text)	Output (Gender)
d_1	iPhone7 is a good mobile	Male
d_2	Battery of this phone is bad	Female
d_3	I am enjoying using iphone7	Male

Example 03 – Treating Gender Identification Problem as Learning Input-Output Function

Available Data

Instance No.	Input (image)	Output (Gender)
d_1		Male
d_2		Female
d_3		Male

Example 04 – Treating Gender Identification Problem as Learning Input-Output Function

Available Data

Instance No.	Input (Audio)	Output (Gender)
d_1	Audio-001.mp4	Male
d_2	Audio-002.mp4	Female
d_3	Audio-003.mp4	Male

Example 05 – Treating Sentiment Analysis Problem as Learning Input-Output Function

Available Data

Instance No.	Input (Text)	Output (Sentiment)
d_1	iPhone7 is a good mobile	Positive
d_2	Battery of this phone is bad	Negative
d_3	I am using iphone7	Neutral

Example 06 – Treating Machine Translation Problem as Learning Input-Output Function

Available Data

Instance No.	Input (Source Language Text)	Output (Target Language Text)
d_1	Allah is One	اللہ ایک ہے
d_2	Understanding is deeper than Love	تفہیم محبت سے زیادہ گہری ہے
d_3	The only thing that is constant in this world is change	اس دنیامیں واحد چیز جو مستقل ہے وہ ہے تبدیلی

Example 07 – Treating Object Detection Problem as Learning Input-Output Function

Available Data

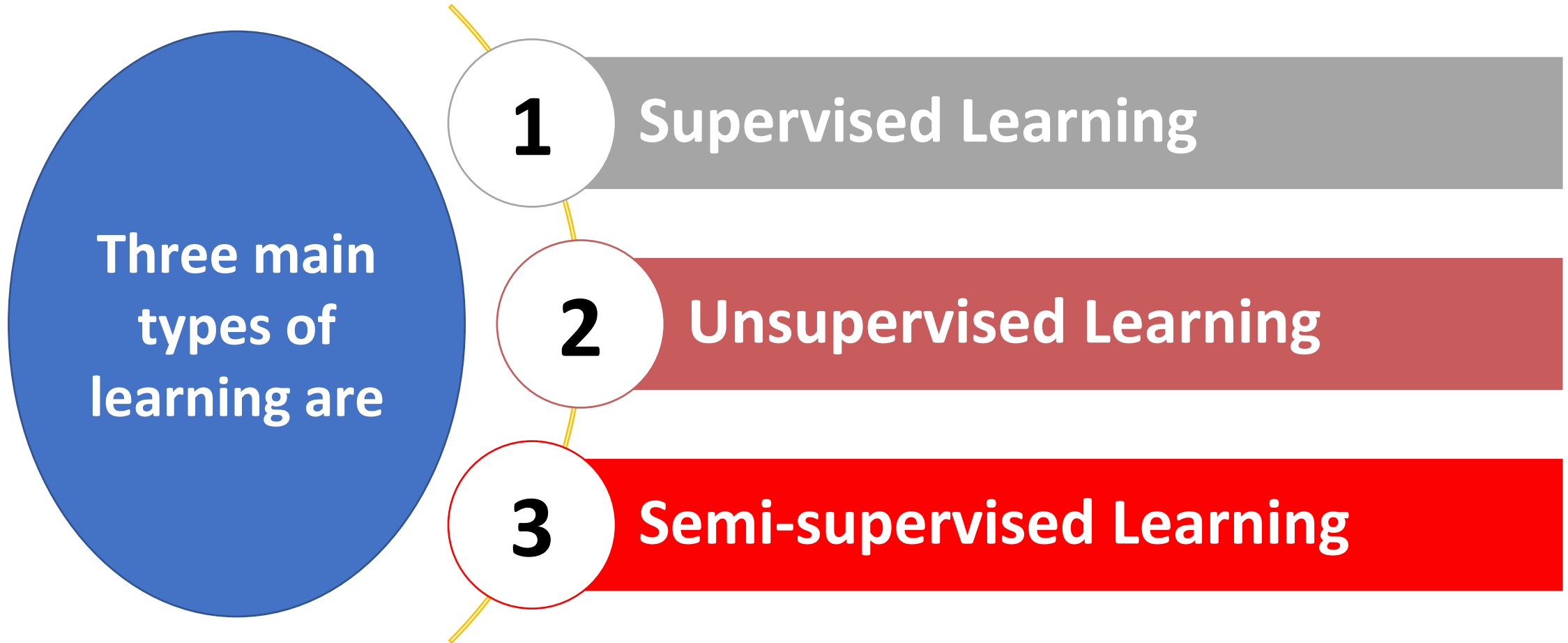
Instance No.	Input (Image)	Output (Object Detection in Image)
d_1		

Example 08 – Treating Natural Language Description Generation from Image Problem as Learning Input-Output Function

Available Data

Instance No.	Input (Image)	Output (Image Description)
d_1		A cup of coffee for breakfast to wake up

Types of Learning



Types of Supervised Learning Cont...

Example

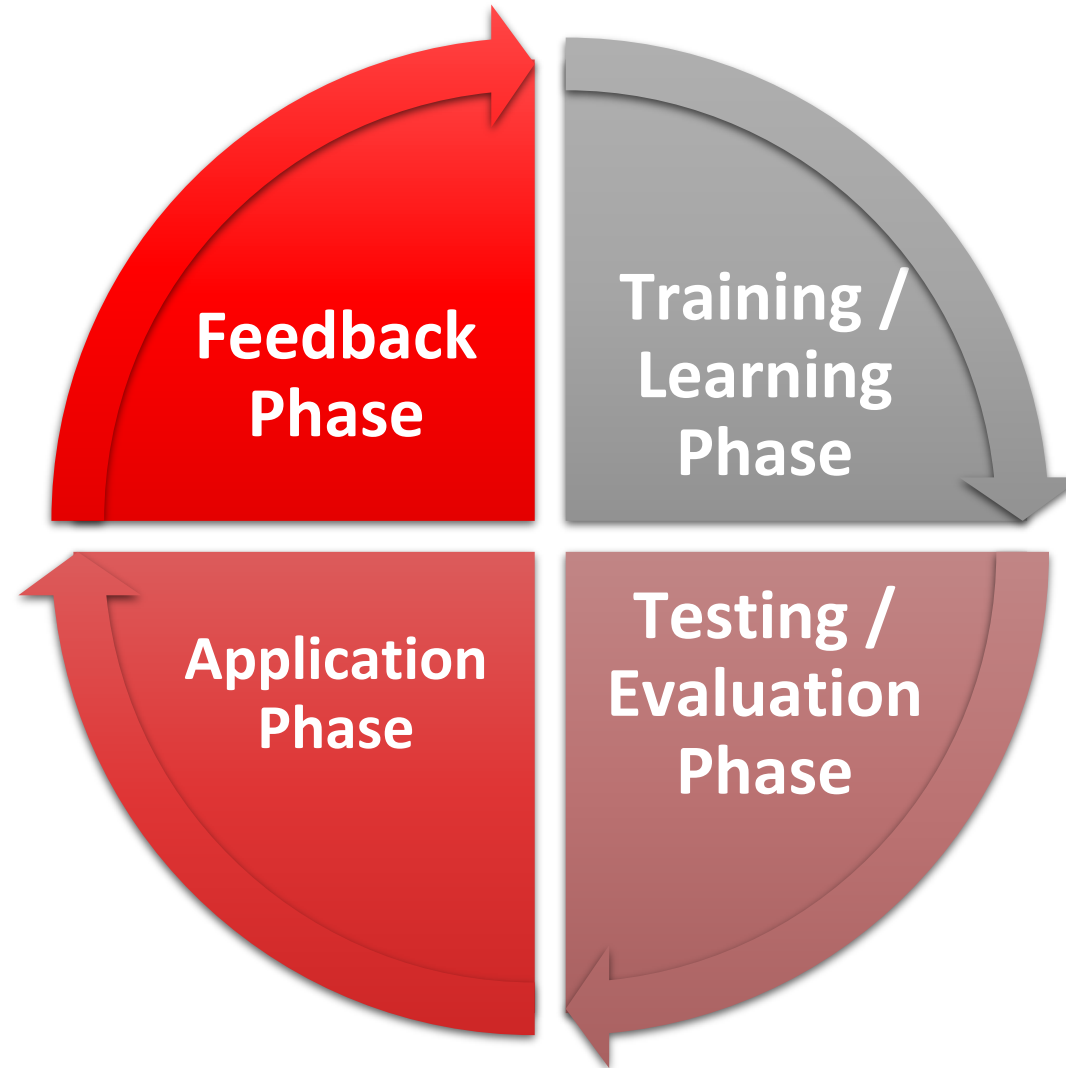
Task: Gender Identification – Annotated Data

Instance No.	Input					Output
	Height	Weight	Hair Length	Beard	Scarf	Gender
d ₁	180.3	196	Bald	Yes	No	Male
d ₂	170.0	120	Long	No	No	Female
d ₃	178.5	200	Short	No	No	Male
d ₄	163.4	110	Medium	No	Yes	Female
d ₅	175.2	220	Short	Yes	No	Male
d ₆	165.0	150	Medium	No	Yes	Female
d ₇	179.1	185	Long	Yes	No	Male
d ₈	160.5	130	Short	No	No	Female
d ₉	177.8	160	Bald	No	No	Male
d ₁₀	161.1	100	Medium	No	No	Female



Learning Cycle

Learning Cycle



NLP Working

